Application of Environmental Condition-Aware Frameworks for 3D Structural RNA Prediction on Riboswitch Data

Janae Moses-Soto
dept. Molecular Medicine
jmmosessoto@usf.edu
Dr. Taseef Rahman
dept. Computer Science
taseefr@usf.edu
[University of South Florida]
Tampa, FL

***Abstract*—The advancement of RNA structure prediction utilizing machine learning must move beyond simply the sequence. RNA can undergo multiple conformations, which can make it difficult to determine structure with existing methods in Structural Biology. In this article, the effect of environmental factors such as conformational state, ligand interactions, pH, crystal growth and differential temperatures on RNA structural prediction in machine learning is evaluated utilizing riboswitch data.**

Google Colab: Final_MLProject

Keywords—3D RNA Structures, machine learning, style, riboswitch, Structural Biology

I. Introduction

Machine Learning to predict the structures of biological macromolecules are becoming more common and useful tools to the field of biology. Models such as AlphaFold2 and RoseTTAFold have been critical in the modeling of protein three-dimensional structure with many successes in research. In comparison, there have not been as many machine learning models that have high-accuracy in predicting three-dimensional RNA structures. There is a gap in research and technology. Contests such as Stanford's RNA 3D Folding competitions and the RNA Puzzle competition have been created to bridge the gap in RNA structure prediction. Results can be submitted to the Critical Assessment of Protein Structure Prediction [CASP]. Current machine learning models are built by a "common strategy" in which models are pretrained on RNA sequences and then fine-tuned for structure prediction using annotated data which includes 2D and 3D structures [1]. This strategy leads to an output of a structure that is visually represented as being static. In the journal article, *Machine Learning Generation of Dynamic Protein Conformational Ensembles,* the author highlights the principle that "biomolecules are inherently dynamic, and there is a pressing need for accurate predictions of dynamic ensembles across multiple functional levels" [2]. This principle does not only apply to proteins, but to RNA as well. Limitations in predicting RNA structures are caused by a lack of experimental data and "RNA's complex conformational landscape, characterized by structural flexibility, formation of multiple states, and a propensity to misfold" [3]. Understanding the structure of RNA is important as "RNA molecules play a crucial role in diverse biological processes. Their functions include gene expression regulation, catalyzing biochemical reactions, and genetic information translation" [1]. There is a gap in the research and principles of current machine learning when applied to RNA structures because the environment is not considered, which is necessary to accurately model a dynamic structure. There is not a model, which can fully replicate the environment and apply environmental changes to the construction of RNA structural modeling. The application of environmental condition aware framework in 3D Structural RNA prediction is explored through the creation of a new riboswitch dataset with environmental, sequence-based, and structural-based features. The riboswitch dataset was used for the development of machine learning framework for the prediction of riboswitch conformational states (Apo, Intermediate, and Holo).

II. Related Work

A. Understanding RNA Stability

It is important to understand the structure of RNA and the environmental conditions that play a role in RNA formation. The journal article, *RNA Stability: A Review of the Role of Structural Features and Environmental Conditions* main focus is "to review the main factors affecting RNA stability and describe the existing approaches to its stabilization, providing science-based recommendations for the storage and handling of RNA samples" [4]. This journal article aids in the understanding of the environmental and chemical conditions affecting RNA such as the ionic environment, post-transcriptional modifications, and features of RNA secondary and tertiary structure.

B. Factors Affecting Stability

The primary structure of RNA is a sequence of nucleotides: Adenine, Guanine, Cytosine, and Uracil which contribute to its stability and function [4]. The authors highlight the mechanism of different environmental conditions affecting RNA stability. A key factor to RNA stability is the 2'OH group in the ribose which acts as a nucleophile and attacks and break the phosphodiester bond near it, especially in high pH environments [4]. It is also mentioned that Ca2+ and other metal ions can affect the environment, noting that "this process is highly dependent on the specific ionic environment and the presence of metal ions that can stabilize the transition state and promote catalysis" [4]. The authors mention that the poly(A) tail of RNA affects stability [4]. However, it is not mentioned how this feature should mostly be considered in eukaryotic mature RNA (mRNA). Modifications of bases by post-transcriptional

modifications such as methylation, pseudouridylation, and uracilyation are mentioned by the authors to affect RNA stability and interactions with proteins [4]. RNA secondary structure is also an important feature to consider, with the authors stating it "serves as the basis for the formation of the complex three-dimensional structure of the molecule" [4]. Secondary structures such as hairpins, loops, and pseudoknots are mentioned as being characteristics in mRNAs, tRNAs, rRNAs, microRNAs, and small-interfering RNAs [4]. It is noted by the authors that eukaryotic mRNAs contain approximately 10-15% nucleotides involved in secondary structures in comparison to some viral mRNA containing approximately 30-40% nucleotides involved in secondary structures [4]. The tertiary structure of RNA does not only include nucleotide sequence as in the primary structure, but it also "is a complex three-dimensional shape formed by interactions between nucleotide bases, the sugar-phosphate chain, and metal ions" [4]. This is significant because the nucleotide sequence alone cannot fully represent the tertiary structure of RNA. Additional tertiary structure elements include trunks, loops, and quadruplex groups [4]. The ions involved in the stability of RNA's tertiary structure are Mg2+, Ca2+, and Mn2+[4]. Additional ions that were mentioned are Zn2+, Fe2+, and Fe3+[4]. It is mentioned by the authors that these ions also play an important role for RNA polymerase, an enzyme responsible for RNA synthesis and Ribonucleases P and H [4]. Understanding the factors that affect RNA structure and stability is critical to accurately modeling in machine learning.

C. Current Experimental Methods to Capture RNA Structure

The current experimental methods used to solve RNA instability in vitro (outside of the cell) was explored in *RNA Stability: A Review of the Role of Structural Features and Environmental Conditions.* The methods in stabilizing RNA include the use of RNA stabilizing reagents, RNA-stabilizing matrices, cryopreservation, lyophilization, RNA-stabilizing nanoparticles, chemical modifications, and RNA chaperones [4]. This information shows experimental approaches by laboratory scientists. Some methods may be expensive [4]. Additionally, the authors make recommendations for RNA storage [4]. Environmental condition aware frameworks in Machine learning that model experimental conditions in vitro (outside of the cell) and compare them to in vivo (inside the cell) can help to bridge the gap between the stability of experimental and real RNA structures.

Experimental methods to capture RNA structure include X-ray crystallography, NMR, and Cryo-EM. Currently, Cryo-EM produces the highest resolution images of RNA structures. However, this equipment is expensive, which makes it inaccessible to some laboratories. Bonilla and Jang mention that 3D modeling of atomic structures can be used to produce cryo-EM maps (3). This is significant to research because the cryo-EM maps are stored in the public repository, Electron Microscopy Data Bank (EMDB). Bonilla and Jang state, "tools for de novo prediction of RNA 3D structure can be used to generate starting models that can be refined using the cryo-EM density, these tools include Alphafold3, RosettaFoldNA, and FARFAR2" (3). This statement by Bonilla and Jang illustrate the use of cryo-EM as experimental data in machine learning to predict RNA structure (3). However, they also point out that cryo-EM maps greater than 4Å need validation experiments to validate the atomic models (3).
The authors, Bonilla and Kang of Challenges, advances, and opportunities in RNA structural biology by Cryo-EM summarize cryo-EM's potential in analyzing RNA with complex structural conformations to include dynamic riboswitches, viral RNAs, and untranslated regions of RNA (3). They discussed the need for more bioengineering approaches for experimental applications. Bioinformatics approaches to cryo-EM map modeling are needed as well, with Bonilla and Kang stating, "due to the moderate resolution of typical cryo-EM maps of RNA, integration of powerful computational tools for RNA 3D modeling are crucial" (3). They also list the need for more computational tools to provide a full evaluation of the RNA states that Cryo-EM produces (3). Cryo-EM maps may prove to be a useful comparison to machine learning RNA 3D predicted structures. Understanding experimental methods to capture RNA structure is crucial in the development of computational tools and machine models to support laboratory workflows.

D. Machine Learning Models for RNA Structure Prediction

There have been many advances and challenges in recent years for machine learning models which predict RNA structure. *Machine Learning in RNA structure Prediction: Advances and Challenges* outline current techniques and areas that prove difficult and are not fully solved when using machine learning as an RNA structure predictor. The authors state, "accurate determination of RNA structures through experimental techniques and computational modeling hold significant importance not only in understanding RNA functions, but also in drug and therapeutic designs" (1). It is mentioned by the authors that there are several types of RNA: mRNA, tRNA, rRNA, ncRNAs which have their own specific functions (1). Experimental techniques such as x-ray crystallography, NMR spectroscopy, Cryo-EM, and SHAPE cannot fully encompass "the growth of biologically significant structures" (1). Zhang, Li, and Chen (1) highlight the need for computational RNA structure predictors in addition to experimental techniques. This article goes into detail about the techniques used in modern Machine Learning models and ongoing research in this field. They go on to mention the current traditional structural learning models for both 2D and 3D structure predictions. Thermodynamic Traditional models for 2D structure predictions that Zhang, Li, and Chen (1) mention include RNAfold, Mfold, RNAstructure, and Vfold2D. A lack of thermodynamic data for RNA structures is a challenge for these models (1). Structural template-based method models listed in *Machine Learning in RNA structure Prediction: Advances and Challenges* are Vfold3D, VfoldLA, 3dRNA, and RNAComposer. An issue for these models is the lack of templates from solved RNA structures available (1). A common occurring challenge is the lack of solved RNA structures. Zhang, Li, and Chen provide many more examples for de novo (new) RNA structure predictor models to include NAST, iFoldRNA, SimRNA as a few examples (1). The challenges that this type of model faces are inefficiency when performing conformational sampling in an atom or coarse-grained 3D space and the lack of accurate scoring or energy functions (1). In comparison, Zhang, Li, and Chen explain how machine learning features such as neural networks and

deep learning is another significant approach to overcome the challenges previously mentioned (1). This approach to machine learning relies on known RNA structural data, without requiring as much knowledge of physics (1). The use of neural networks also allows models, "to capture more intricate relationships and patterns between sequence features and structures" (1). Zhang, Li, and Chen mention models E2Efold-3D, RoseTTAFoldNA, and DRfold as utilizing deep neural networks (1). Unfortunately, these models have not been as successful in RNA structural prediction as AlphaFold2 in protein structure prediction (1). It can be argued that a different or modified approach is needed to predict RNA structure due to RNA functional differences compared to proteins.

E. Inputs and Outputs of Neural Networks

The inputs for 2D modern RNA structural prediction are likely to include RNA sequences and Multiple Sequence Alignments (MSAs) (1). Zhang, Li, and Chen provide an example of a model that incorporates this called SPOT-RNA2 (1). SPOT-RNA2 includes 2D features from the Multiple Sequence Alignment (MSA) such as the Position Specific Score Matrix (PSSM) and 2D Direct Coupling Analysis (DCA) to input into the neural network (1). Zhang, Li, and Chen suggest that the addition of more structural features such as "thermodynamic and covariation-predicted features" will improve the accuracy of RNA structural prediction (1). Modern machine learning models to predict 3D structures use 2D predicted structures, MSAs, or both as inputs (1). Accurate 2D structural prediction plays a significant role in ensuring 3D structural predictions (1). Multiple Sequence Alignments (MSAs) offer evolutionary insight (1). Machine learning models trained with this data according to Zhang, Li, and Chen, "exhibit a superior performance in predicting RNA structures" (1). The cons of using MSAs are constructing effective MSAs with a lack of RNA sequence data and their time-consuming for large biological datasets (1). The outputs in 2D modern RNA structural prediction may include the 2D structure, the 2D contact map, or the folding scores (1). The authors point out that the 2D contact map is the most flexible output for further analysis such as in thermodynamic studies (1). Zhang, Li, and Chen state, "Considering the intrinsic conformational heterogeneity of RNA's, the ability to predict alternative structures is crucial to the understanding of RNA function" (1). This statement highlights the need for current machine learning models to predict multiple structures in a multiple structure ensemble instead of a singular static structure. In comparison, the outputs in 3D RNA structural prediction may vary (1). Model examples listed by Zhang, Li, and Chen which produce an output of multiple geometric constraints are DeepFoldRNA and trRosettaRNA (1). Other models such as E2Efold-3D, Drfold, and RoseTTAFoldNA produce an output of the computation of RNA 3D structures (1). It is mentioned that models like the ones listed above can improve the speed accuracy of RNA 3D structural prediction (1). Accuracy of RNA 3D structural prediction using current machine models is still limited by their outputs which may be "a single, static RNA 3D structure" (1). Zhang, Li, and Chen explain how this limitation does not allow 3D RNA structural prediction to match biological functionality (1). They make suggestions to solve this problem and implement multiple structure ensemble by altering input MSAs, exploring energy landscapes, and the addition of generative models (1). Suggestions such as these align closely with the need for machine learning models with environmental condition-aware frameworks, which can accurately model the conditions RNA structure undergoes in different environments.

F. Evaluation of RNAlight for Environmental Features

RNAlight is a machine learning model based on LightGBM which predicts subcellular localization of RNA types by identifying specific nucleotide k-mers which characterize subcellular localization [5]. This model was originally trained on mRNAs and lncRNAs, but has been extended to include snRNAs, snoRNAs, and a variety of circular RNA transcripts [5]. The authors, G.H. Yuan, Y. Wang, G.H. Wang, and L. Yang emphasize the importance in understanding subcellular localization to RNA function [5]. Subcellular localization is also important to RNA structure if you apply the structure-function paradigm. RNAlight was developed to fill the gap in machine modeling development for a subcellular localization predictor of different RNA types [5] Shapley Additive exPlanations (Tree SHAP) and k-mer assembly in RNAlight were found to discover more insight into distinct sequence features and their associated RNA binding proteins (RBPs) in subcellular localizations of mRNA and lncRNA [5]. RNAlight could prove to be a useful tool for analyzing and recreating the environmental conditions which RNA undergoes.

The first step in developing RNAlight was data collection to create a combined data set which included data from reported data sets containing lnCRNA and/or mRNA subcellular localization [5]. Datasets for lncRNA subcellular localization included data from LncATLAS, CeFra-seq, and APEX-Seq [5]. Next, these were filtered using specific algorithms to include only nuclear and cytoplasmic RNA [5] Using the LncAtlas number of lncRNA and cell lines, an equation was created to measure nuclear localization versus cytoplasmic localization [5]. The equation created was (CN – RCImean) in which RCImean is the mean of the cytoplasmic and/or nuclear concentration spanning thirteen cell lines [5].
lncRNAs with a score of (CN – RCImean) < -2 (n = 1857) were considered to have nuclear localization, while lncRNAs with a score of (CN – RCImean) > 0 (n = 1440) were considered to have cytoplasmic localization [5]. The $CytoFPKM$ equation: $CR = CytoFPKM + NucFPKM$ was created to separate and extract nuclear and cytoplasmic RNA in the CeFra-seq data set [5]. The lncRNAs in the APEX-Seq dataset which had at least one component in the nucleus, nucleolus, lamina, and nuclear pore were categorized as being nuclear RNAs if they had alog2(fold change) ≥ 0.75 [5]. In comparison, the lncRNAs in the APEX-Seq dataset which had at least one component in the cytoplasm, endoplasmic reticulum (ER) membrane, ER lumen, and outer mitochondrial membrane were categorized as being cytoplasmic RNAs if they had a log2(fold change) ≥ 0.75[5]. Additionally, the developers, G.H. Yuan, Y. Wang, G.H. Wang, and L. Yang collected "a totally independent dataset containing lncRNA and mRNA subcellular localization information from Halo-seq in HeLa cell line" to evaluate RNAlight against other models [5]. In further paragraphs, the authors discuss the featurization of RNA primary sequences and the featurization of RNA secondary structures utilizing Vienna RNAfold and the bpRNAalgorithm [5]. After featurization, RNAlight was

constructed using the LightGBM Python package to train RNAlight on the k-mer frequency matrix and labeled subcellular localizations from training data [5]. The optimal hyperparameters for LightGBM were selected using RandomizedSearchCV [5]. Next, the construction of Support Vector Machine (SVM) and logistical regression models were created using the Python scikit package [5]. To construct the deep learning-based models CNN, RNN, and a hybrid CNN + RNN were applied utilizing Tensorflow backend in Python [5]. Lastly, model performance was evaluated. Further construction and fine tuning related to the identification of k-mers from the nucleus or cytoplasm, the identification of RBP-associated motifs related to RNA subcellular localization, calculation of light score, and selection of circular RNAs was conducted as a base for erformance [5]. The methodology behind the development of RNAlight is clearly documented.

RNAlight outperformed deep-learning models in predicting RNA subcellular localization [5]. It was shown to be an accurate model for various RNA types [5]. The authors mention that RNAlight can differentiate RNA cirlces with exon or intron origins [5]. RNAlight also "showed the inclusion of intron sequences in lncRNA, mRNA, and circRNA could substantially change their subcellullar localization from cytoplasm to nucleus" [5]. Theoretically, this change in subcellular localization could affect RNAs conformational structures. The authors hypothesize that extracting only sequence features may be "sufficient enough to achieve RNA subcellular localization prediction [5]. The authors list training RNAlight on additional datasets which include more features such as RNA secondary structures or chemical modifications may better train it for more complex and dynamic subcellular localization [5].

G. Language Model-Based Deep Learning Approach – RhoFold⁺

The Accurate RNA 3D structure prediction using a language model-based deep learning approach by T. Shen, Z. Hu, Sun, S., and et al, details the creation of RhoFold+ as an RNA 3D structural predictor [6]. The authors "present RhoFold+, an RNA language model-based deep learning method that accurately predicts 3D structures of single-chain RNAs from sequences" [6]. The lack of solved 3D RNA structures in the Protein Data Bank (PDB) is mentioned by the authors [6]. This further emphasizes the challenge in structural biology in understanding the mechanism of RNA folding. According to the authors, the existing methods for RNA 3D structure are categorized into either template-based modeling or de novo prediction approaches [6]. MSA-based approaches are evaluated to be successful in predicting 3D RNA structures, but time-consuming to construct [6]. DRFold is used as an example of a model that does not use MSAs and the input is a single sequence [6]. DRFold creates and predicts 3D RNA structures based off the 2D RNA structural prediction [6]. This method is highly accurate and not as time intensive [6]. RhoFold+ is presented as being a de novo predictor which leverages speed and accuracy for RNA 3D structural prediction [6]. It is an improved model from RhoFold which adds "integration of MSAs and other features to improve performance" [6]. RhoFold+ was created with framework for only single chain RNAs [6].

To incorporate evolutionary data into the RhoFold+ pipeline and use it as an additional input, the software Infernal and rMSA were used to construct MSAs [6]. Infernal was used to search for all possible conserved secondary structures and cross-referenced against RNA sequence databases using rMSA, as a tool to search through RNA sequence databases [6]. Next, The MSAs that were constructed were reduced to 256 sequences as the maximum limit during the training stage due to memory constraints [6]. After, 256 MSAs were also "either randomly selected or chosen through clustering and then fed into RhoFold+" [6] A technique that was applied was clustering utilizing conserved sequences from secondary structures and embeddings from their pretrained RNA language model [6]. The top 256 MSAs were selected as inputs and the expected output structure would be an "optimal model" [6]. RhoFold+ was adapted from RhoFold. The authors provide an overview of the RNA-FM language which is a transformer model with framework in RhoFold+ that was built from the (BERT) transformer language model [6]. The RNA-FM was trained using a large noncoding RNA (ncRNA) data set from RNAcentral [6]. The data was preprocessed before training by replacing Thymine (T) with Uracil (U) and removing duplicate sequences [6]. RhoFold+'s RNA-FM consists of 12 transformer-encoder blocks [6]. The authors explain that this model can take "raw sequential tokens as input, mapping each nucleotide into a 640-dimensional vector via an embedding layer, forming an L x 640 embedding matrix" [5]. The matrix is then passed through every encoder block plus a softmax layer to predict tokens which include sixteen nucleotides and four functional identifiers [6]. The RNA-FM was trained on eight 80 GBA100 GPUs for a month [5]. They set a limit of 1,024 sequence length for faster training [6]. An additional curated synthetic data set was used with origins from PDB [6]. The model was retrained using 25% of data from PDB and 75% of synthetic data [6]. The confidence measure, pLDDT was used to exclude pLDDT scores of less than 0.7% for pseudo-labeled data [6]. In RhoFold+ framework the coordinates of RNA are modeled as the atom positions (C4', C1', N1/N9) and the torsion angles α, β, γ, ω using IPA [6]. Structural prediction undergoes a "recycling process" and only stops when pLDDT converges [6]. Feature processing and structure prediction loss is also detailed in this article [6]. Feature processing describes the process the machine learning model undergoes to get the predicted 3D coordinates [6]. The loss of function is considered at all levels of RNA structure leading up to 3D, which includes 1D, 2D, and 3D [6]. At the two-dimensional level, distance loss, secondary structure loss, and cross-entropy loss is considered [6]. At the three-dimensional level, the model calculates the sensitivity to the 3D error utilizing frame aligned point error loss (FAPE), secondary structure constraint loss, and clash violation loss to improve model performance when constructing RNA 3D structures [6]. The loss is constant if there is a difference in the predicted structure versus the actual structure by rotation and translation [6]. Overall, the model evaluates loss and confidence metrics to prevent overfitting and provide accuracy [6].

RhoFold+ utilizes an RNA language model-based deep learning method to successfully create accurate RNA 3D structures within 0.14 seconds [6]. It does not use the traditional method of MSA searching which can be a time-consuming process [6]. It has been evaluated to maintain a high accuracy rate against RNA-puzzle targets from the RNA puzzle competition and naturally occurring RNAs from the Critical Assessment of Protein Structure Prediction 15 (CASP15) [6]. The authors prove through evaluation that

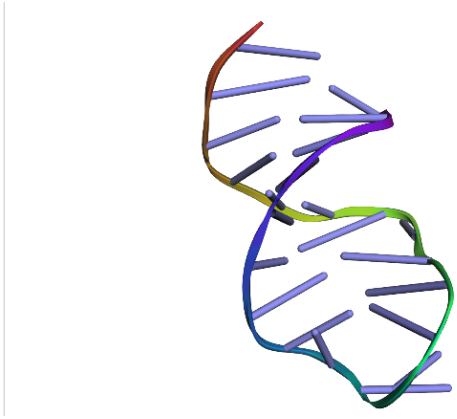
"RhoFold+ does not overfit its training set" and maintains a high accuracy to unseen RNA structures [6]. It is important to note that when evaluating RhoFold+ structures against CASP15, synthetic targets were excluded [6]. RhoFold+ was evaluated against six naturally occurring RNA target structures from CASP15 [6]. This is significant because the excluded structures can be classified as in vitro RNA meaning synthesis is experimental [6]. The developers of RhoFold+ believed that the synthetic structures were out of scope of what RhoFold+ was trained on [6]. Additionally, RhoFold+ had a higher accuracy than RhoFold with an average r.m.s.d. of ~1Å when evaluated against CASP15 structures [6]. A performance issue that RhoFold+ could improve upon is the alignment of helices in the 3D RNA structure [6]. The developers believe that this error is caused because RNA does not exist in only one conformation but shifts through conformations and has dynamic and flexible junctions in its structure [6]. RhoFold+ also outperformed AlphaFold3 and RoseTTAFold2NA [6]. AlphaFold3 and RoseTTAFold2NA are designed differently than RhoFold+ which only takes single RNA sequences versus RNA-protein complexes [6]. This is interesting because the designs of AlphaFold3 and RoseTTAFold2NA are meant to encompass the full scope of interactions which RNA undergoes during folding; however, they do not perform as well on simple RNA. The question then arises as to how a machine learning model can accurately represent both large and complex RNA, and simple single chain RNA? Additional comprehensive testing was performed using that RhoFold+ on diverse types of RNA with developers seeing a positive performance on tRNA and miRNA, but "struggling with introns and riboswitches" [6]. Introns and riboswitches typically characterize noncoding RNA which can be more complex. Solutions for improvement in this area may lie in the preprocessing selection of data. Particularly, in the selection of the 256 top MSAs that were chosen as input features and fed into RhoFold+ [6] How were the top 256 MSAs chosen? Does the selection guarantee diversity? Lastly, the developers addressed other limitations of RhoFold+ to include difficulties in predicting conformations of the same RNA, large and complex RNA structures, RNA-ligand models, RNA models with diverse environmental conditions, and synthetic RNA structures [6].

A. Reproduction and Evaluation of RhoFold+ for 3D RNA Structure Prediction

To fully evaluate the top-performing machine learning model for 3D RNA structure prediction, RhoFold+ was "technically" reproduced with available resources. RhoFold+ was chosen for reproduction of an RNA 3D structural prediction machine learning model because it is one of the most accurate and efficient models currently used. Additionally, the reproduction of RhoFold+ may help to identify more areas of improvement that can be made in RNA structural prediction. It does not utilize traditional Multiple Sequence Alignment Method searching [6]. It is outperforming models such as AlphaFold3 and RoseTTAFold2NA [6]. It can successfully create accurate RNA 3D structures within 0.14 seconds and "RhoFold does not overfit its training set" [6]. Additionally, it has been evaluated against RNA structures from the Critical Assessment of Protein Structure Prediction 15 (CASP15) [6]. It is important to note that synthetic RNA structures were not included when evaluating RhoFold+ against CASP15 as the developers believed it was out of RhoFold+ scope at the time [6]. A performance issue that was considered in evaluation was RhoFold+ struggled to align the helices in the 3D RNA structure [6]. It was believed that this issue stemmed from the different conformations that RNA can exist at [6]. Other limitations of RhoFold+ that were addressed are its difficulties predicting conformations of the same RNA, large and complex RNA, RNA-ligand models, RNA models with varied environmental conditions and synthetic RNA [6].

Google Colab was used as the platform for the reproduction of RhoFold+ because its Colab Pro plan is free for students and educators. RhoFold+ is a large language model and requires a significant amount of parallel processing to provide an output. Google Colab makes it easier to explore these models without needing access to NVIDIA GPU support or a cluster. During this reproduction, the connected runtime was set at A100 GPU. RhoFold+ was not fully "implemented from scratch" as this would be impossible with the time frame, resources, and technical expertise it would take to create. The article, Accurate RNA 3D structure prediction using a language model-based deep learning approach outlines RhoFold+ architecture from scratch [6]. However, the environment that RhoFold+ utilizes was reproduced. Existing libraries such as PyTorch, BioPython, and OpenMM were used to implement RhoFold+ pipeline and rebuild the prediction engine. First, condacolab was installed and RhoFold+ was cloned. A new environment was created using mamba. The contents of the directory was checked to verify the RhoFold+ directory was properly cloned. Next, a MPLBACKEND was set, rhofold_pretrained_params.pt was checked, and the input file 3mei A.fasta was set, and the inference was set for a single input sequence. The generated PDB files were listed. py3Dmol was installed to visualize the 3D RNA structure output from RhoFold+. The results were downloaded and saved. PDBParser and Superimposer were imported from BioPython. PDBParser was used to read the PDB, while Superimposer is used to rotate/translate and superimpose a set of atoms onto each other to minimize Root Mean Square Deviation (RMSD) (7). The model was loaded and the file was read. The reference structure was downloaded from the Protein Data Bank (PDB). The PDB ID is 3MEI. Next, only the RNA backbone atoms from both the model and reference structures are counted by filtering for Phosphorous and Carbon 4 atoms (C4). After printing the number of backbone atoms for both the reference and the model structure, the two structures were superimposed and the RMSD was calculated. For comparison, the MSA file (A3M) for 3MEI was also used as another structure prediction. According to T. Shen, Z. Hu, Sun, S., et al in Accurate RNA 3D structure prediction using a language model-based deep learning approach, "higher MSA quality correlated with better performance" [6]. Therefore, the MSA structure should be more accurate than the singular sequence structure of 3MEI [6]. The inference for the 3MEI MSA file was run and the new predicted model was loaded. The RMSD was also calculated for the 3MEI MSA file. The RMSD was also verified using the educational PyMol Version 3.1 by Schrödinger (8) for the single sequence model.

Figure 1: 3MEI Chain A py3Dmol-2.5.4 predicted from Reproduced RhoFold+



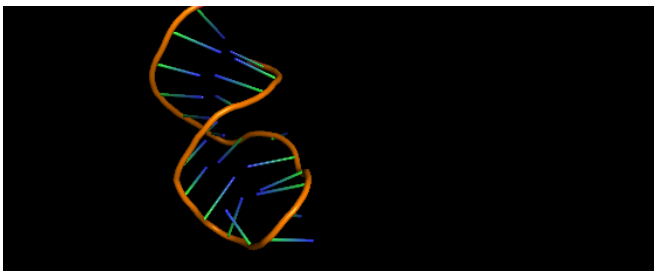


Figure 2: 3MEI Chain A Single Sequence Prediction Reproduced RhoFold+ PyMol Version 3.1 by Schrödinger (Educational Use Only)

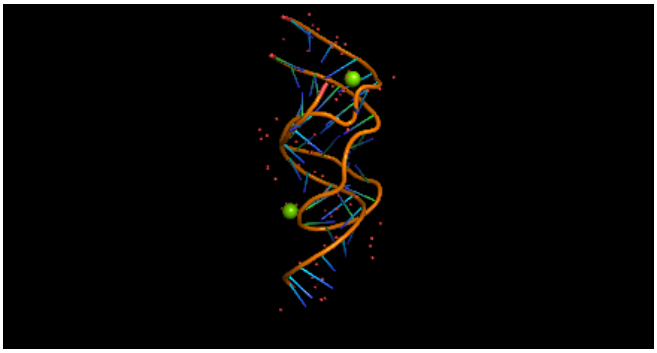


Figure 3: 3MEI Chain A Alignment for Predicted Reproduced RhoFold+ and PDB Structure PyMol Version 3.1 by Schrödinger (Educational Use Only)

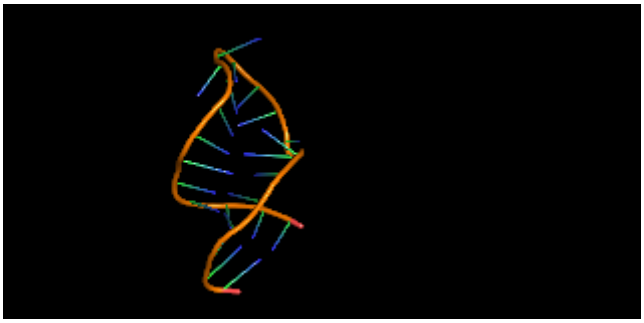


Figure 4: Figure 2: 3MEI Chain A MSA Prediction Reproduced RhoFold+ PyMol Version 3.1 by Schrödinger (Educational Use Only)

The accuracy and confidence of the reproduced RhoFold+ is not as high as the original model. This may be due to limited RAM/GPU, lack of MSA, and limited seedlings. It was run on the single-sequence interference and then, the MSA interference. It is noted by the developers of RhoFold+ that the single-sequence function is less accurate than the MSA interference mode (8). The calculated RSMD in Google Colab was 18.083Ä for the single-sequence interference. The calculated RSMD in Google Colab was 19.425Ä. The predicted structures from the reproduction of RhoFold+ were also aligned utilizing pyMol Version 3.1 by Schrödinger (Educational Use Only) (8). The calculated RSMD for single-sequence interference using pyMol was 17.156 (22 to 22 atoms). The calculated RSMD for the MSA was 9.014 (367 to 367 atoms). The RMSD for the RhoFold+ online platform single-sequence interference was approximately 7 when aligned with the reference PDB structure in pyMol. As reference, the RhoFold+ model can achieve a RSMD as low as 2.384 on 3meiA [6]. Overall, a reproduction of RhoFold+ was executed and can be improved in the future to predict more accurate structures. The link to the RhoFold^{+} can be found on Google Colab: RhoFold Reproduction

G. Initial Project Proposal

After the evaluation of the challenges in machine learning for RNA structure predictors, it is obvious there is a need for a model which accurately predicts conformational ensembles based on the environment. The easiest solution being to modify the parameters of a highly accurate 3D RNA predictor such as RhoFold+ and adjust to environmental conditions than to create a completely new model. This could be done by using other models such as RNAlight to provide more insight into subcellular localization. A hybrid-approach to the RNA structural problem is needed. This would require the restructuring of a data set into a new data set to train the model from multiple open-source repositories. Additionally, there must be considerations to efficiency and memory limitations, as well as modifying the framework to combine models. To complete this challenge, there is a need for strong biological knowledge and machine learning background. The creation of an accurate predictor for RNA folding in vivo and in vitro could advance structural biology as we know it.

III. Methodology

After technically reproducing RhoFold^{+}, it became clear that it would not be feasible to attempt to modify the parameters of a deep learning RNA predictor such as RhoFold+ to account for environmental conditions or create a completely new model. RhoFold requires a significant amount of GPU to run for efficient and accurate predictions. However, combining a strong visualization prediction model (RhoFold^{+}), a subcellular environmental prediction model (RNAlight), and a classification model into one model is significant future work in structural biology. With respect to the scope of this project, time, and resources an XGBoost Classifier model with a two-staged classification system was trained on riboswitch data. This model was utilized in the development of a machine learning framework for the prediction of riboswitch conformational states (Apo, Holo, Intermediate) from environmental features, sequence-based

features, and structural features. A new riboswitch data set was created. Riboswitches were chosen as a representative of RNA due to their flexibility and switch in conformations based on the "binding of a cognate ligand to the ligand-sensing domain" [9]. Riboswitches are categorized by the name of the ligand they bind to and provide a varied dataset to train and test a machine learning model against [9]. Most riboswitch classes are prokaryotic apart from the only Eukaryotic riboswitch class discovered, TPP [9]. It is still an area of exploration, and riboswitches are used in machine learning models due to their relatively small size which uses less storage.

IV. Experiment

Dr. Taseef Rahman acted as a mentor during this project and assisted in the creation of the methodology by creating an outline of how this project could be complete. This document can be found in the references [10]. Riboswitch data organized by class was downloaded from https://riboswitch.ribocentre.org [11]. The riboswitch classes fall under categories such as RNA derivatives, RNA precursors, signaling molecules, elemental ions, amino acids, sugars, cofactors, and others [11]. The largest known and validated representative group of riboswitches is cofactors to include 20 different types of cofactors [11]. The smallest known representative group of riboswitches is sugars which only includes GlcN6P [11]. The full list of riboswitches was downloaded from the database. Representative riboswitches were taken from each group besides sugars because it only includes one type of sugar-riboswitch. This downloaded dataset provided the PDB ID, riboswitch class, and ligand information [11]. Next, PDB files for the selected riboswitch needed to be downloaded. The riboswitch PDB files can be found at https://www.rcsb.org [12]. This could be completed manually from the website or by utilizing a program. The expected output was a folder of .cif files [10]. This was completed by running a python script in the terminal, "Extract_PDB.py". The actual outout was 10 folders which were named as Adenine, cdiGMPI_ID, Fluoride_id, FMN_id, Glutamine I_id, GlutamineII_id, PreQ1_id, SAMI_id, and THF_id. Approximately 123 pdb files was downloaded and organized into their appropriate folder. After, downloading the riboswitch files, the environmental conditions needed to be extracted from the headers (variable names) in the metadata [10]. Some examples of key fields to extract were "exptl_crystal_grow.pdx_details", "_exptl_crystal_grow.pH", "_diffrn.ambient_tempz", and "_pdbx_number_of_molecules", and etc. [10]. Some of these fields were inconsistent or missing entirely [10]. After the environmental features were extracted, the conformational states (Apo, Holo, and Intermediate) of each riboswitch sample were labeled. These labels were reviewed and confirmed by reading the additional experimental information or finding this information in the title on https://www.rcsb.org. Data extracted had to undergo a cleaning process and missing data had to be entered as NA [10]. This process can be completed manually or via a program. Extracting and cleaning the data set was the most intensive and time-consuming task. Computing sequence and structural features were completed immediately after the data extraction and cleaning. In this step, the RNA sequence was pulled from riboswitch_PDBdata.xlsx and the length, GC content, dinucleotide frequencies, and 3mer frequencies. The command line was used to install Vienna RNAfold which provided calculations for the minimum free energy and structural element counts of each riboswitch in my dataset [10]. An optional but recommended step which Dr. Taseef suggested was to calculate the root-mean-square deviation in PyMol for riboswitches that have both Apo and Holo structures [10]. This step would have been very valuable and meaningful but in the interest of time was skipped. However, this step should be included in future studies. Majority of the data mining and processing occurred locally in the terminal and was later uploaded into Google Colab. Overall, the riboswitch data set created contains 123 riboswitches with three categories of features: environmental, sequence-based, and structure-based [10].

A. Machine Learning Workflow

Google Colab was used as the platform for the reproduction. The necessary libraries were installed and imported which includes scikit-learn, xgboost, iblearn pandas, NumPy, seaborn, and sklearn. The Google drive was mounted. The riboswitch data set was loaded. The riboswitch data set was adjusted after uploading in the drive to add the crystal growth details because they did not initially get extracted in the original riboswitch data set. This feature had to undergo more extensive cleaning and the data was normalized. The data was reloaded to include this feature. Features which were created from existing features were ion concentrations (in mM) for Magnesium, Potassium, Sodium, Manganese, and Cobalt. Additionally, pH was measured. Enthalpy and transition width was also calculated for each riboswitch as additional features. The features were categorized into environmental, sequence and structural to make "all_features." The target was defined as the Conformational State. After, the data frame was cleaned again to ensure the removal of missing values. Another ligand feature was created has_Barium or SAM to capture more specific interactions that might be biologically significant. Another feature created was "GC_content_x_Length". After all of the new features were added they were re-encoded, Hyperparameter tuned, and LOGO Cross-Validation. The accuracy rate of each riboswitch class was evaluated and average feature importance analyzed. This occurred multiple times and many visualizations were created. The riboswitch class FMN was left out of the training data initially because it has multiple conformational states which I believed would be good to test on. Leaving one class out of the data set was to assess model generalization. Sample_weights and custom prediction thresholds were used in training to enhance recall for minority classes. Due to the model's difficulty predicting FMN, a Binary Classifier for FMN detection was created to predict whether a riboswitch or sample is a FMN riboswitches. FMN Overall, a Main classifier for the majority of riboswitch was created and FMN classifier. In the future, a two-stage prediction function could be integrated and FMN evaluated more closely as the reasoning it is an exception to the Main classifier.

B. Datasets, Settings, and Metrics

All datasets can be found on the Google Colab and at the link below: https://github.com/jmmoses99/ML-Project. As previously stated, the data was downloaded from https://riboswitch.ribocentre.org. The PDB files are located on the PDB website: https://www.rcsb.org. All of the data

used in this project is publicly available. This machine learning model was trained utilizing the CPU setting and the H100 GPU setting. This model does not require a strong runtime like RhoFold+ because it is a simple classifier, meant only to prove the relevance of environmental features in RNA structural prediction and the need for an accurate hybrid-machine learning model. The metrics used were accuracy, precision, recall, and F1 score, and a balanced accuracy score. A matrices which would have made the project more meaningful is RMSD from PyMol.

V. Results (Tables, Figures, Comparisons)

The results can be found in the Google Colab, but are also included in this section.

```
--- FINAL POLISHED REPORTS ---

========================================
 FINAL REPORT: Adenine
========================================
              precision    recall  f1-score   support

         Apo       1.00      1.00      1.00         1
        Holo       1.00      0.70      0.82        10
Intermediate       0.25      1.00      0.40         1

    accuracy                           0.75        12
   macro avg       0.75      0.90      0.74        12
weighted avg       0.94      0.75      0.80        12


========================================
 FINAL REPORT: Fluoride_id
========================================
              precision    recall  f1-score   support

         Apo       0.00      0.00      0.00         0
        Holo       1.00      1.00      1.00         4
Intermediate       0.00      0.00      0.00         0

    accuracy                           1.00         4
   macro avg       0.33      0.33      0.33         4
weighted avg       1.00      1.00      1.00         4


========================================
 FINAL REPORT: GlutamineII_id
========================================
              precision    recall  f1-score   support

         Apo       0.00      0.00      0.00         0
        Holo       1.00      1.00      1.00         1
Intermediate       0.00      0.00      0.00         0

    accuracy                           1.00         1
   macro avg       0.33      0.33      0.33         1
weighted avg       1.00      1.00      1.00         1


========================================
 FINAL REPORT: PreQI_id
========================================
              precision    recall  f1-score   support

         Apo       1.00      1.00      1.00         1
        Holo       1.00      1.00      1.00        10
Intermediate       0.00      0.00      0.00         0

    accuracy                           1.00        11
   macro avg       0.67      0.67      0.67        11
weighted avg       1.00      1.00      1.00        11


========================================
 FINAL REPORT: SAMI_id
========================================
              precision    recall  f1-score   support

         Apo       1.00      1.00      1.00         3
        Holo       1.00      1.00      1.00         6
Intermediate       0.00      0.00      0.00         0

    accuracy                           1.00         9
   macro avg       0.67      0.67      0.67         9
weighted avg       1.00      1.00      1.00         9


========================================
 FINAL REPORT: THF_id
========================================
              precision    recall  f1-score   support

         Apo       1.00      1.00      1.00         1
        Holo       1.00      1.00      1.00         9
Intermediate       0.00      0.00      0.00         0

    accuracy                           1.00        10
   macro avg       0.67      0.67      0.67        10
weighted avg       1.00      1.00      1.00        10


========================================
 FINAL REPORT: TPP_id
```

Figure 5: Result Report for Riboswitch Classes

```
Finished Fold: Adenine          | Accuracy: 0.83
Finished Fold: Fluoride_id      | Accuracy: 1.00
Finished Fold: GlutamineII_id   | Accuracy: 1.00
Finished Fold: PreQI_id         | Accuracy: 0.91
Finished Fold: SAMI_id          | Accuracy: 0.67
Finished Fold: THF_id           | Accuracy: 0.90
Finished Fold: TPP_id           | Accuracy: 1.00
Finished Fold: cdiGMPI_id       | Accuracy: 0.82
```

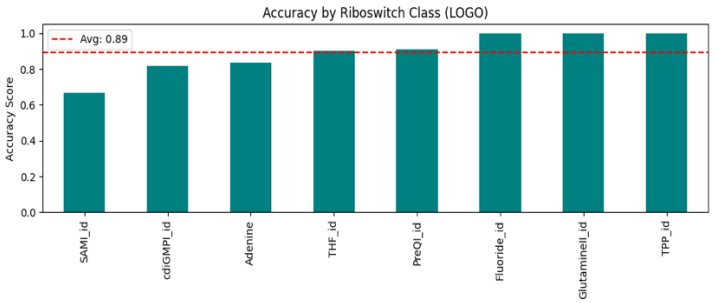


Figure 6: Accuracy by Riboswitch Class

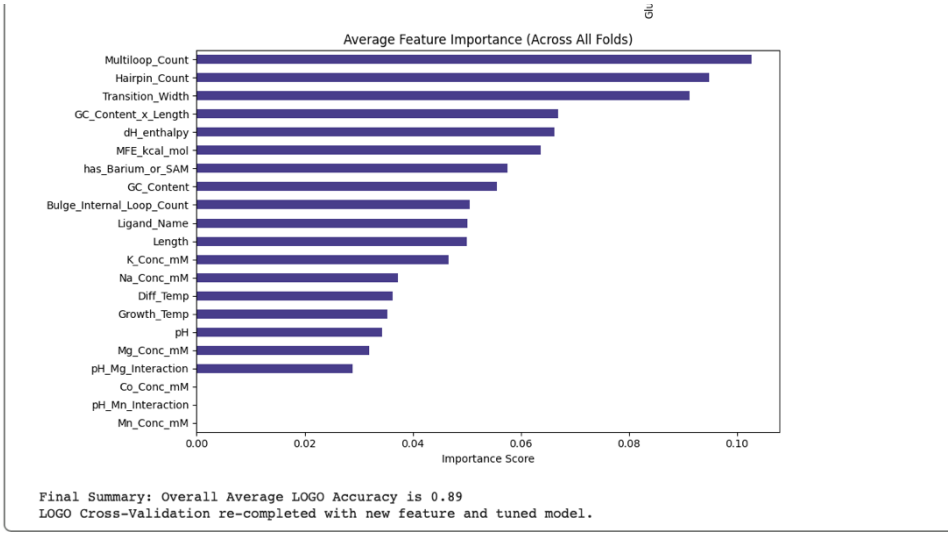


Figure 7: Average Feature Importance (Across All Folds)

```
Distribution of 'Conformational State' in df_clean:
Conformational State
Holo            76
Apo              8
Intermediate     7
Name: count, dtype: int64

Distribution of 'Folder' (Riboswitch Classes) in df_clean:
Folder
TPP_id          19
FMN_id          14
Adenine         12
PreQI_id        11
cdiGMPI_id      11
THF_id          10
SAMI_id          9
Fluoride_id      4
GlutamineII_id   1
Name: count, dtype: int64
```

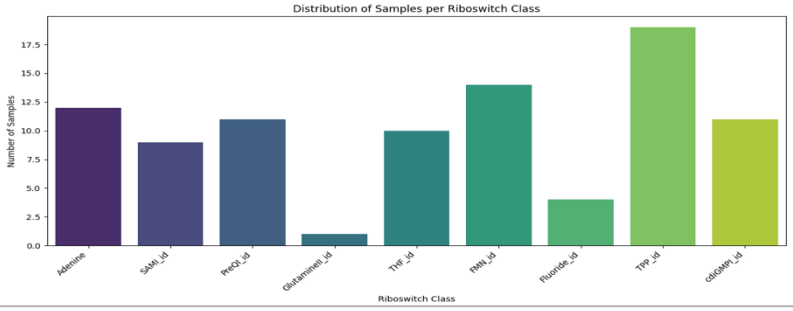


Figure 8: Distribution of Samples per Riboswitch Class

Figure 9: Distribution of Growth_Temp for SAMI_id vs Other Classes

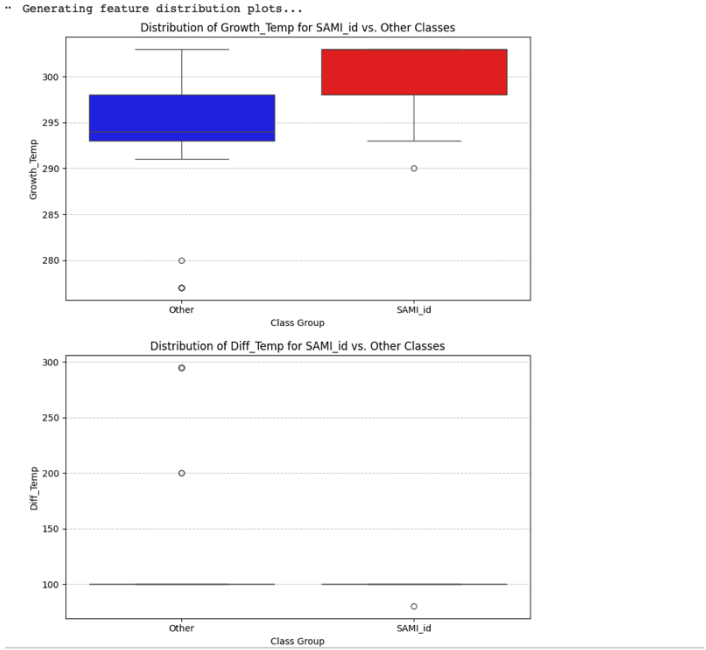
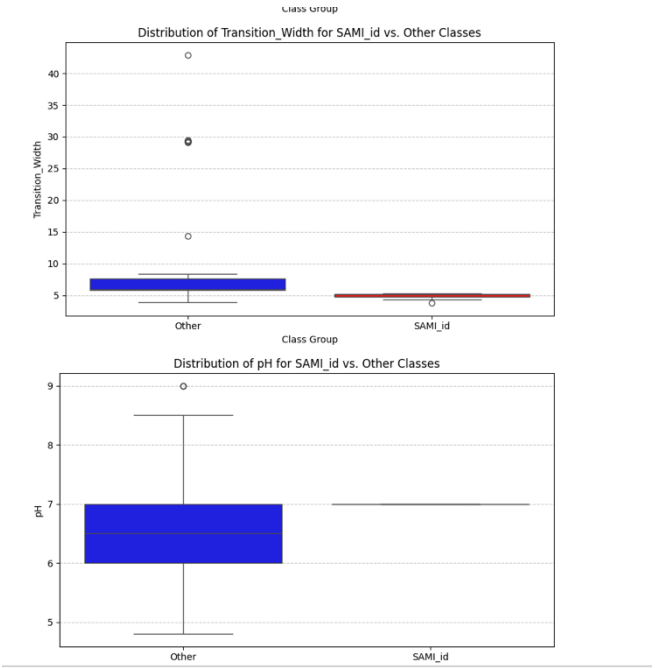


Figure 10: Distribution of Transition_width and pH for SAMI_id vs Other Classes



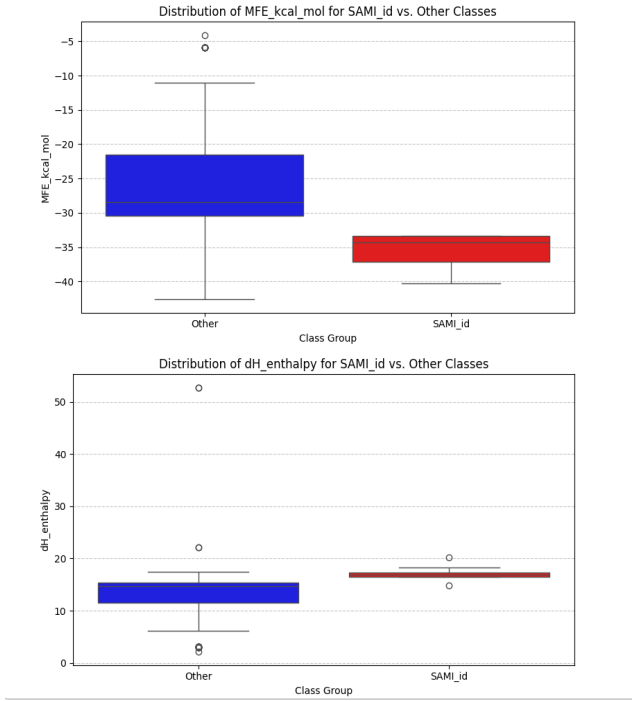


Figure 8: Distribution of MFE_kcal_mol and dH_enthalpy for SAMI_id vs Other Classes

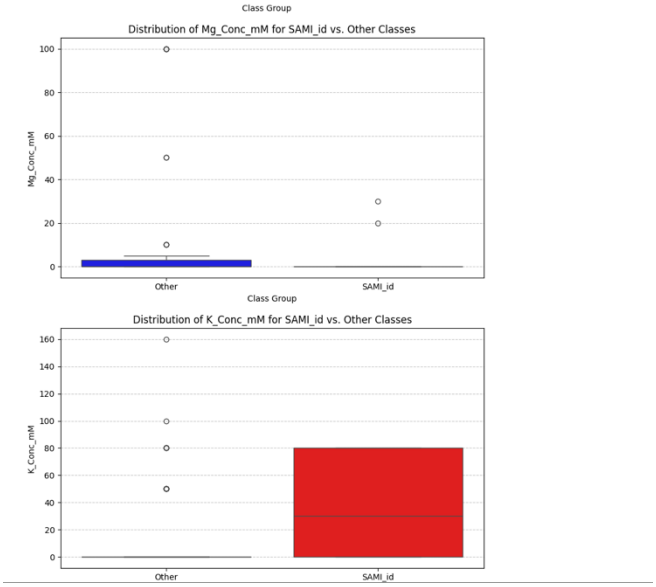


Figure 11: Distribution of Mg_con and K_conc for SAMI_id vs Other Classes

Figure 12: Distribution of GC_Content for SAMI vs Other Classes

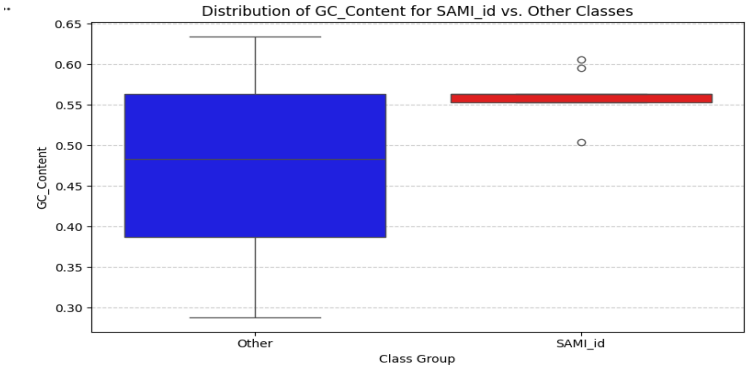
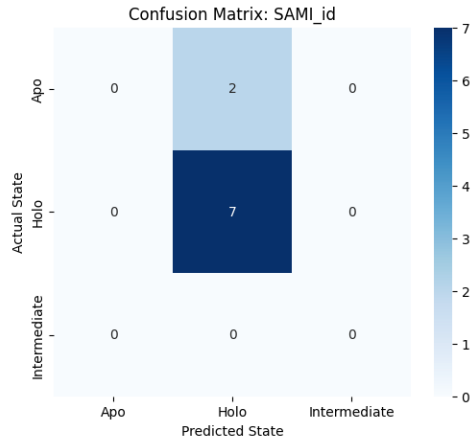


Figure 13: Confusion Matrix Worst Performing Fold

```
··· Analyzing Worst Performing Fold: SAMI_id (Accuracy: 0.67)
```




```
Classification Report for SAMI_id:
              precision    recall  f1-score   support

        Apo       0.00      0.00      0.00         2
       Holo       0.78      1.00      0.88         7
Intermediate      0.00      0.00      0.00         0

   accuracy                           0.78         9
  macro avg       0.26      0.33      0.29         9
weighted avg      0.60      0.78      0.68         9
```

```
··· Descriptive statistics for 'GC_Content_x_Length':

FMN_id:
count    14.000000
mean     52.644450
std       0.632544
min      52.001600
25%      52.001600
50%      53.002500
75%      53.002500
max      53.995200
Name: GC_Content_x_Length, dtype: float64

Apo_Training (Excluding FMN_id):
count     7.000000
mean     44.713857
std      16.829105
min      15.998400
25%      36.502700
50%      52.000800
75%      55.996300
max      59.999800
Name: GC_Content_x_Length, dtype: float64
```

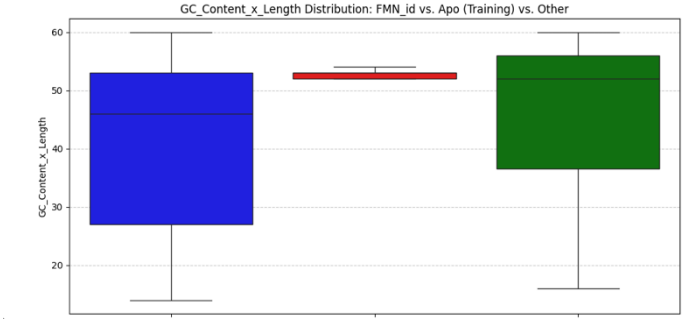


Figure 14: GC_Content_x_Length Distribution FMN_id vs Others

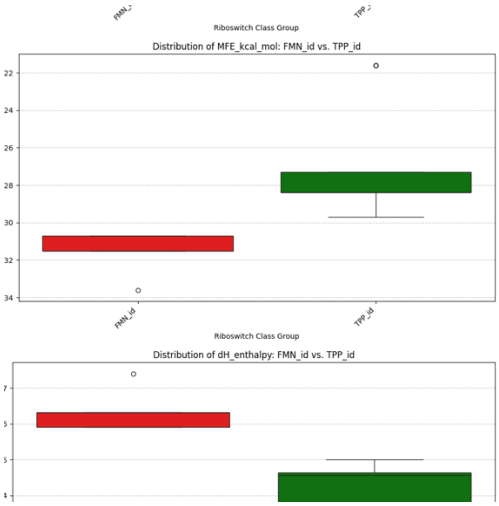


Figure 15: Distribution of MFE_Kcal_mol and dH_Enthalpy FMN_id vs TPP_id

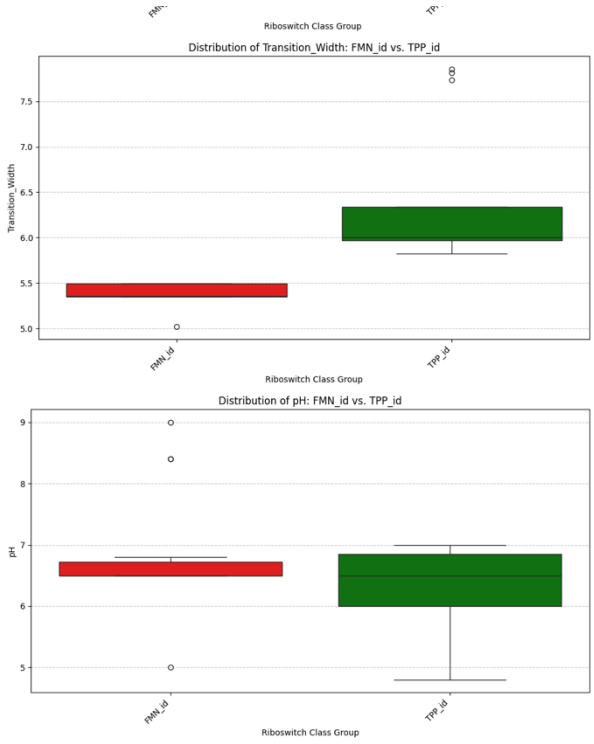


Figure 16: Distribution of Transition_Width and pH FMN_id vs TPP_ID

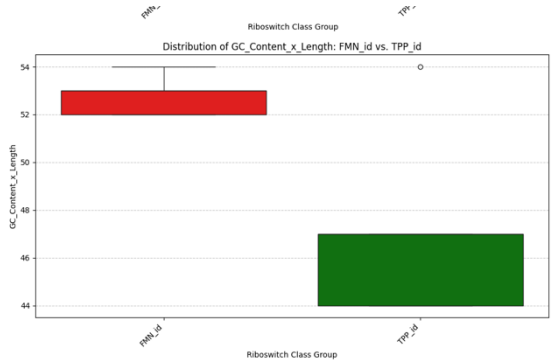


Figure 17: Distribution of GC_Content_x_Length FM vs TPP_id

VI. Discussion

Exploring the application of environmental condition aware framework in RNA structural prediction was analyzed by creating a XGBoost Classifier machine learning model with a two-staged classification system which was trained using a diverse riboswitch dataset. This model predicts which conformational state (Apo, Intermediate, or Halo) a riboswitch structure will be in based upon environmental, sequence-based, and structure-based features. The overall performance of this two-staged system was 0.8791. It was also developed using hyperparameter tuning using GridSearchCV with StratifiedKFold. This model had some challenges accurately predicting the SAMI and FMN riboswitch class. SAMI shows differences in the amount of potassium ions it interacts with and growth temperature. The FMN riboswitch was left out of training but was still struggling with accurate predictions of conformational states. The FMN riboswitch may be the exception because this class contained a greater number of samples with diverse conformational states. In Figure 17, the GC Content compared with the length of the FMN class is significantly different compared to TPP's values. The energy landscapes of the conformational states which the FMN exists as seem to be different from TPP as well. The creation of a new riboswitch dataset which includes these features and is publicly available is a significant contribution to the fields of structural biology and machine learning.

VII. Conclusion and Future Work

This project showcases the importance for RNA structural prediction utilizing machine learning models to move beyond solely using sequence as an input for 3D RNA predictors. The environmental, sequence-based, and structural-based features move a prediction from being static to dynamic. A dynamic structural prediction will be the most accurate to a RNA's native state. A model with strong and accurate visualization, subcellular environmental predictions, and classification of conformational state an RNA exists in would be the next "AlphaFold2" for RNA. Overall, this project provided a blueprint for future 3D RNA structural models. It provided enriching features, dynamic state prediction, and leveraged thermodynamic features. It was a great starting point for a bioinformatician trying to understand the capabilities and applications of machine learning in biology. It provided great practice for data mining and cleaning complex and "messy" data. This model has not been tested against the entire data set of riboswitches to see the accuracy of different types even within the same class. This is an area for future analysis. The incorporation of environmental features to a visualization predictor is an area worth exploring. The data set created from this project could be reused. The future applications are limitless.

References

[1] S. Zhang, J. Li, and S. Chen, "Machine learning in RNA structure prediction: Advances and challenges," Biophysical Journal, vol. 123, no. 17, pp. 2647-2657, September 2024.

[2] L.-E. Zhang, S. Barethiya, E. Nordquist, and J. Chen, " Machine Learning Generation of Dynamic Protein Conformational Ensembles, in Molecules 28, no. 10, 4047, 2023.

[3] S. L. Bonilla and K. Jang, "Challenges, advances, and opportunities in RNA structural biology by Cryo-EM" in Current Opinion in Structural Biology, vol. 88, p. 102894, October 2024

[4] I. V. Kornienko, O. Y. Aramova, A. A. Tishenko, D. V. Rudoy, and M. L. Chikindas, "RNA Stability: A Review of the Role of Structural Features and Environmental Conditions," in Molecules, 29, no. 24, 5978, 2024.

[5] G.H. Yuan, Y. Wang, G.H. Wang, and L. Yang,"RNAlight: a machine learning model to identify nucleotide features determining RNA subcellular localization," in Briefings in Bioinformatics, vol. 24, no. 1, January 2023.

[6] T. Shen, Z. Hu, Sun, S., et al., "Accurate RNA 3D structure prediction using a language model-based deep learning approach," in Nature Methods, 21, pp. 2287-2298, November 2024.

[7] The Biopython Contributeors "Bio.PDB.Superimposer Module," in biopython.org, 1999-2020.

[8] PyMOL
The PyMOL Molecular Graphics System, Version 3.1 Schrödinger, LLC.

[9] J.R. Stagno, Y.-X. Wang, "Riboswitch Mechanisms for Regulation of P1 Helix Stability," Int J Mol Sci. 25(19):10682, October 4,2025 doi: 10.3390/ijms251910682

[10] https://github.com/jmmoses99/ML-Project/blob/main/riboswitch_dataset_documentation.pdf

[11] F. Bu, X. Lin, W.Liao, et. al, "Ribocentre-switch: a database of riboswitches," Nucleic Acids Res. gkad891(2023), https://pubmed.ncbi.nlm.nih.gov/37855663/

[12] H.M. Berman, J. Westbrook, Z. Feng, G. Gilliland, T.N. Bhat, H. Weissig, T.N. Shindyalov, P.E. Bourne, "The Protein Data Bank, Nucleic Acids (2000) Research 28: 235-242 https://doi.org/10.1093/nar/28.1.235.